





Project
01
02
03
04
05

TacChat

Empowering User Generation and Accessible Sharing of Tactile Graphics for the Blind and Visually Impaired

Team Project
2025/07 - 2025/10
Future Lab, Tsinghua University
Jiao Yang Team
Research Project

Role: **Co-first Author**
Paper: Submitted to **IMWUT**
Contribution of mine:
System workflow design, Experimental design, Data analysis

Research

While refreshable tactile graphics help BLV users understand images, few studies explore **their needs as tactile content creators.**

Instead of being passive consumers, BLV individuals wish to **proactively express themselves**, and even **make contribution to the overall connection and well-being of BLV community.**



Interviews in local uiniversity for blind students

Final work



a) 3D printing pen



b) Drawing on canvas



c) RTD device





Our RTD device for tactile graphics and text display was showcased at the **"Accessibility for All: Building an Inclusive Future"** exhibition, held at the United Nations Office at Geneva from June 16 to 21, 2025.

How To Use

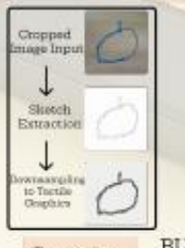
The authoring workflow of TacChat is organized in three main steps—Explore and Author, Convert and Modify, Share and Communicate—to support BLV individuals in creating and sharing tactile graphics.

Explore and Draw



BLV authors sketch on the canvas with a 3D printing pen.

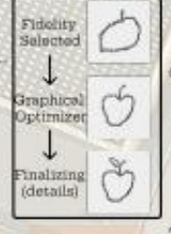
Convert and Modify



Converting ...



BLV authors check the draft on RTD and do some modification.



Modifying ...

Share and Communicate



The graphics are sent to readers' RTD for sharing and communication.

Testing and Conclusion



Through TacChat, BLV users gained a sense of **agency and engagement** in creating, reading, and sharing tactile content, enhancing **both their understanding and expression of information**, while also **fostering social interaction, community building, and envisioning future possibilities of tactile media.**

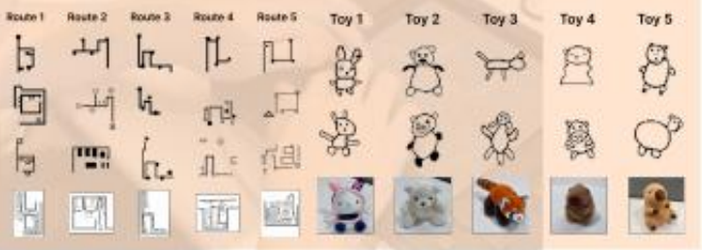


"For visually impaired individuals like us, there's not much we can do in daily life. This essentially **gives us a brand new skill.**"

"We're used to being passive recipients, but this kind of authoring and sharing **gives me a strong sense of participation.**"

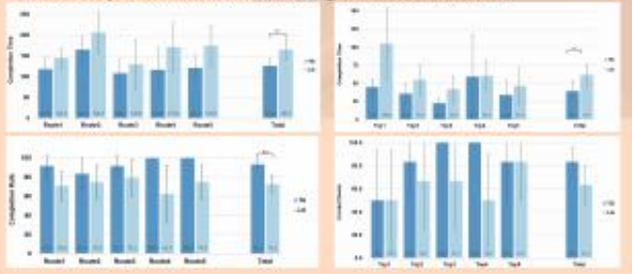
"I feel it's like sighted people can see with their eyes, while we can touch with our hands. **It's kind of equal exchange.**"

The subject's work



TacChat significantly improved **both accuracy and efficiency**

In the Route Navigation Experiment, users achieved a **93.3% completion rate** and finished tasks **25% faster** than the verbal-only group. In the Toy Assembly Experiment, their decision time was **reduced by one-third**, with a **20% higher correctness rate**.



TacChat received **a Good overall UEQ-S rating** (mean = 1.43), indicating that BLV participants found it **engaging, user-friendly, and supportive of tactile authoring and sharing.**

